

EDUCATION

UNIVERSITY OF MICHIGAN	Ann Arbor, US
PhD in Mechanical Engineering	Sept. 2024- June 2028
UNIVERSITY OF WASHINGTON	Seattle, US
MSc in Materials&Science Engineering	Sept. 2022- June 2024
UNIVERSITY OF NOTTINGHAM, NINGBO CAMPUS	Zhejiang, China
BEng (Hons) Mechanical Engineering	Sept. 2018- June 2022

ACADEMIC RESEARCH

SCALABLE SOLID-STATE BATTERY PROCESSING & MULTIPHYSICS MODELING - June. 2025 – Current

- Developed multiphysics models in COMSOL to simulate spray deposition, improving coating uniformity
- Conducting parametric studies (velocity, geometry) to optimize deposition performance
- Designing scalable manufacturing workflows for solid-state battery components, integrating dry processing
- Fabricated solid state battery cell which achieved **93 % Coulombic efficiency**

PULTRUSION OF ELECTRICAL VECHICLE COMPONENTS MADE OF LOW-COST CARBON FIBER COMPOSITES - Dec. 2024 – Current

- Developed side-pole crash simulations for electric vehicle structures using LS-DYNA
- Evaluated and optimized composite battery module separators for lightweight and impact safety performance
- Reduced pulling resistance by ~60 %** through design optimization of composite architecture and processing parameters
- Developing machine learning-based methods to optimize fiber distribution in composite structures

ROBOT-BASED ADDITIVE MANUFACTURING OF LEGO™-TYPE MODULAR MOLDS FOR WIND BLADES - Oct. 2024 – Dec. 2024

- Conducted temperature profiling of wind turbine molds using thermocouples, capturing thermal history across **multiple locations (±4 °C variation)**
- Evaluated vacuum integrity and surface deformation, **maintaining vacuum levels below ~10 kPa** and reducing **surface deviation to within ~0.5 mm**
- Fabricated glass fiber/epoxy composite components using 3D-printed molds, achieving consistent part quality and dimensional accuracy

CARBON FIBER PREPREG RECYCLING - Oct. 2022 – June. 2024

- Fabricated recycled carbon fiber composites via bulk molding compound processing, achieving consistent laminate quality across multiple samples
- Conducted digital image correlation to capture mechanical response and applied convolutional neaural network to predict failure
- Developed Python-based machine learning models using datasets of ~100 samples to correlate processing conditions with composite performance

THERMOMECHANICAL BEHAVIOR AND FORMABILITY OF AEROSPACE THERMOPLASTIC PREPREGS - June 2023 - Sep. 2023

- Characterized viscoelastic behavior of thermoplastic prepregs using DMA (stress relaxation and thermal ramp tests)
- Fabricated thin-ply composite laminates via hot pressing under controlled processing conditions
- Evaluated thermomechanical response and bending behavior using DMA-based 4-point bending tests

MULTIPHYSICS MODELING OF WING-STRUCTURED FLEXIBLE SENSORS - June 2022 - Oct. 2022

- Developed finite element models in ABAQUS to simulate the mechanical behavior of wing-structured flexible sensors
- Investigated hyperelastic deformation mechanisms and stress distribution under different loading conditions
- Improved model accuracy through validation and refinement of boundary conditions and material properties

SUSTAINABLE COMPOSITE MATERIALS DESIGN USING RECYCLED CARBON FIBER - June 2020 - Oct. 2022

- Designed and fabricated recycled and hybrid fiber composites for high-performance and sustainable applications
- Integrated natural fibers, fire-retardant (FR) systems, and functional coatings to tailor mechanical, thermal, and electrical properties
- Improved flame retardancy (~1 % increase in LOI)** while maintaining structural performance
- Quantified performance trade-offs, **including 5–10% reduction in impact strength and ~12 % variation in interlaminar shear strength**

MECHANICAL DESIGN FOR MOBILE AND MECHANICAL SYSTEMS - Feb. 2020 - Oct. 2022

- Designed a lightweight gearbox in SolidWorks with gear and shaft calculations
- Developed a Raspberry Pi-based mobile platform with control algorithms (C/MATLAB)
- Enabled basic autonomous movement and obstacle detection
- Fabricated components using machining processes (drilling, milling, cutting)

INTERNSHIP

BUREAU OF INDUSTRIAL INFORMATION TECHNOLOGY - Jan. 2020 - Feb. 2020

- Analyzed the technology and manufacture data to evaluate the tendency of technological development
- Participated in the industrial park field investigation and attended the international cooperation projects

SKILLS

Simulation & Multiphysics Modeling:

ABAQUS (UMAT, thermo-chemical-mechanical modeling), COMSOL Multiphysics (electrostatics, particle tracing), LS-DYNA

Programming & Data Analysis:

Python (machine learning, data-driven modeling), MATLAB, C

Composites Manufacturing & Processing:

Vacuum infusion, hot pressing, hand layup, bulk molding compound (BMC), additive manufacturing

Mechanical & Thermomechanical Testing:

Universal testing machine, drop tower impact, dynamic mechanical analysis (DMA), differential scanning calorimetry (DSC)

Characterization & Measurement:

Digital image correlation (DIC), scanning electron microscopy (SEM), laser profilometry

HONOR

Outstanding Participant in FoSE Elite Project; the First Prize in Online Yangming Cultural Study and Knowledge Contest; Communication and Marketing Campaign Volunteer; AIESEC Volunteer

PUBLICATION

- Wang, Luohaoran, Jacob Harris, Ali Zolali, Elias Shakour, and Mihaela Banu. "Beyond Homogeneity: Fiber Architecture Effects on the Structural Performance of Pultruded Thermoset Composites". SAMPE 2026(2026).
- Wang, Luohaoran, Jacob Harris, Steven Mamolo, Sangharsha Gharat, Ali Zolali, Alan Taub, and Mihaela Banu. "A Theory-Guided Machine Learning and Molecular Dynamics Approach for Characterizing Fast-Curing Polyurethane Systems." Polymers 18, no. 6 (2026): 679.
- Singh, Abhishek, Luohaoran Wang, Abdul Sayeed Khan, Aren Vardhan Pilli, and Mihaela Banu. "Enabling On-Demand Aerospace Component Manufacturing: Topology Optimization of GE Engine Bracket and Fabrication Using Metal FFF." In ASME International Mechanical Engineering Congress and Exposition, vol. 89336, p. V002T03A013. American Society of Mechanical Engineers, 2025.
- Wang, Luohaoran, Jacob Harris, Bhavana Komaraju, and Mihaela Banu. "Crash Simulation Analysis for Optimization of the Carbon Fiber Polymer Composite Pultruded Battery Separators." In International Manufacturing Science and Engineering Conference, vol. 89022, p. V002T13A005. American Society of Mechanical Engineers, 2025.
- Xu, Jin; Feng, Hanfang; Cao, Jinwei; Zhang, Binran; Bao, Wandu; Wang, Luohaoran; Yin, Yimin; Li, Huayang; Zhu, Guang. "A Wing-Structured Sensor with Multiple Mechanical Stimuli Differentiation Capabilities toward Multifunctional Applications". Nano Energy, 2023.

CONFERENCE

- Society for the Advancement of Material and Process Engineering Conference - **April 27 - 30, 2026**
- Manufacturing Science and Engineering Conference - **June 23 – 27, 2025**
- Advanced Materials and Manufacturing Technologies Office Stakeholder Engagement Event - **Dec. 10-11, 2024**
- American Society for Composites (ASC) 38th Annual Technical Conference - **Sep. 17- 20, 2023**